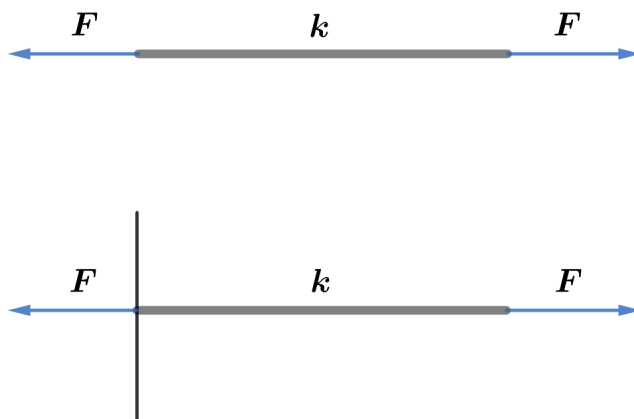


2011 F=ma Exam: Problem 9

Kevin S. Huang

Note that pulling on both ends of the spring is equivalent to the usual scenario of pulling on the spring in one direction with the other end attached to a wall since the wall exerts the same force to keep the spring at rest.



From the definition of the spring constant, the displacement of the free end of the spring is

$$x = \frac{F}{k}$$

which is also the change in length of the spring. Thus, we have

$$L' = L + x = L + \frac{F}{k} = 2\text{ m} + \frac{3\text{ N}}{10\text{ N/m}} = 2.3\text{ m}$$

so the answer is C.